



CHAPTER

Installation

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4.1 Delivery and unloading

TODO: no document describing crane/forklift unloading, container weight, or lifting-point locations was found. The container general-arrangement drawings (2308CA1/2308CA2, appendix B) likely show lifting points but have not yet been reviewed for this purpose.



Figure 4.1. Suggested figure: photograph or diagram of the container being lifted/unloaded on site, with lifting points marked, once the procedure is confirmed.

4.2 Site requirements

The standard unit is a single ISO shipping container (chapter 3), 20 ft/40 ft, High-Cube or Standard. If the optional pretreatment add-on is included, the site needs space for a **second** container alongside it — in HydroVolta's documented BASF pilot, the pretreatment unit used a 10 ft container. The site must provide a water source (the pilot drew from a cooling-tower basin via a submersible pump) and space for a drain tank. TODO: ambient temperature range, ground/ foundation/leveling requirements and clearances beyond what the pilot layout implies were not found in a dedicated site-requirements document.

4.3 Positioning the container

Where the optional pretreatment container is also fitted: from the piping documented for commissioning (section 4.4), its filtered-water outlet (2") must be within pipe-run distance of the main unit's feed inlet (2"), and both containers' 3" drain outlets should be positioned so they can be tee'd together to a shared drain tank.

4.4 Water connections

Standard unit:

- Feed inlet: 2" (M) — feed water must already meet the stack's feedwater envelope (chapter 10) before connection.
- Product-water outlet: 2" (M).
- Drain outlet: 3" (M).
- Fit O-rings and PTFE tape at every threaded joint.

If the optional pretreatment container is also fitted (chapter 8) — these connections apply only to that add-on, not to the standard unit above:

- Pretreatment feed inlet 2½" (M) from the raw water source (in the pilot, a submersible pump rated 4.5 m³/h at 6 bar); filtered-water outlet 2" (M), connected to the standard unit's feed inlet above; drain outlet 3" (M).
- Pump-to-pretreatment feed line: PVCu, 60 mm, terminated 2½" (F).
- Pretreatment-to-unit line: PVCu, 40 mm.
- Both containers' 3" (M) drain outlets combine via a tee into one line to the shared drain tank.
- Key in-line pretreatment components: an ultrafiltration unit (AKKIM APM 80) via 2" Victaulic couplings, and disc filters (AZUD DF100/DF50/DF20) via 2" (F) 63 mm transition pieces.

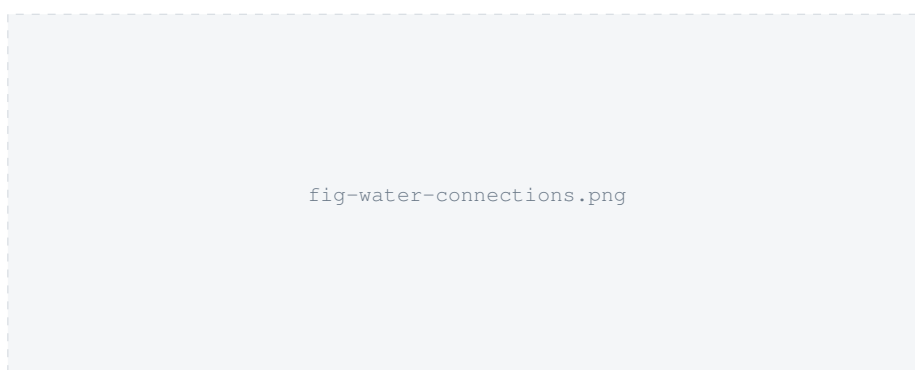


Figure 4.2. Suggested figure: a labeled diagram of the pretreatment/treatment container pair showing every hydraulic connection point (feed, filtered-water, product, drain) with its size and thread type.

4.5 Electrical connection

The control cabinet is IP55-rated. The figures most likely to apply to the **standard unit** come from a separate electrical schematic series for a **treatment container** under the BASF pilot project (folder `2_Electrical/0_Treatment container`, multiple dated AutoCAD Electrical `.dsn` revisions) — TODO: this series was located but not opened (CAD format); a human with AutoCAD Electrical should open the latest revision for the breaker/RCD sizing. The only numeric figures actually extracted so far are from the *pretreatment* container's schematic instead (3×400 V AC, 3P+N+PE per IEC 60445, max. 25 A, 24 V DC control bus for valves/I/O, sheet E1/ document I2) — since pretreatment is an optional add-on (chapters 3 and 8), **do not assume these values apply to the standard single-container unit** until the treatment-container schematic above is actually opened and checked.

4.6 Network and telemetry connection

See chapter 19 for the router, remote-access gateway and cloud dashboard used for network/telemetry connectivity.

4.7 Installation checklist

1. Confirm all hydraulic connections listed in section 4.4 are complete: feed inlet/product outlet/drain (standard unit), plus — only if the optional pretreatment container was ordered — its own feed/outlet/drain and the combined drain-to-tank tee.
2. Confirm electrical supply matches the cabinet's rated voltage/current (chapter 14) before energizing.
3. Confirm network/telemetry connection is established (chapter 19).